Full Aspect Integration as Paper 3's Second Architectural Claim

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Attribution

This work derives from and formalizes the inter-Self coordination architecture introduced in "Inter-Self Coordination via Shared Substrate / Full Aspect Integration" (Li, April 2026), the third paper in the CKS theory series following "Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems" (Li, April 2026) and "The Instinct/Reasoning Separation Outside the Model" (Li, April 2026).

Abstract

Paper 3 of the Coordination Knowledge Substrate (CKS) theory series names six architectural claims. This note formalizes the second: that **Full Aspect Integration (FAI)** is the canonical operation over the shared substrate Paper 3's first claim establishes. Claiming FAI as canonical is not merely a naming decision — it is an architectural commitment about the granularity, default merge posture, cardinality, and exchange scope of inter-Self coordination. The claim specifies that the unit of exchange is the aspect inherited from Paper 2; that the architectural default is full merge of contributed aspects within the shared substrate; that cardinality is n-ary; that three pattern variants govern how merge is configured under human authority; and that exchange is bounded to DNA-layer and action-layer content, with LLM weights and instinct-layer content excluded by architectural commitment rather than by governance configuration. This note states Claim 2 with precision, unpacks each constituent property, situates FAI as the inter-Self analog of Paper 2's mating primitive, and maps seven derived sub-commitments (D1.06–D1.12) as independently claimable items for downstream Phase D1 notes. An operational test closes the note.

1. The Role of Claim 2 in Paper 3's Architecture

Paper 3 builds on a cross-claim spine: the extension of substrate-mediated coordination across the inter-Self perimeter under unified human governance. Each of Paper 3's six claims either establishes an architectural primitive the spine requires, specifies the canonical operation over it, or articulates what the construction composes to at population scope. Claim 1 (§4 of Paper 3) establishes the architectural object — the shared substrate, temporarily constructed for an interaction, perimeter-spanning across distinct Selves' home governance perimeters, carrying all six Paper 1 commitments within scope. Claim 2 specifies what operates over that object.

The specification is not optional and not decomposable into separate architectural decisions. Naming an exchange unit without specifying merge default leaves the architecture open to implementations that operate at the wrong granularity. Naming merge default without specifying exchange scope leaves the architecture open to implementations that exchange instinct-layer content or LLM weights — a violation of the trilogy's foundational instinct/reasoning separation. Naming pattern variants without anchoring them to Paper 2's mating variants leaves them open to claims of novelty that the inheritance record forecloses. Claim 2 bundles the conjunction: aspect-as-unit, full-merge-as-default, n-ary cardinality, three variants, and reasoning-layer-bounded exchange. The conjunction is the architectural commitment; each component is a severable sub-commitment.

2. Claim 2 Stated Precisely

Paper 3 Claim 2: Full Aspect Integration (FAI) is the canonical operation over the shared substrate. FAI has the following properties:

Unit of exchange: The aspect from Paper 2. When a Self contributes content to the shared substrate during an FAI event, it contributes aspects. Contributing an aspect surfaces the aspect's constituent cells, DNA-layer content, and action-layer content into the shared substrate.

N-ary cardinality: Any number of Selves can participate in one FAI event. Two-Self FAI is the simplest instance. One-Self self-derived combination is not an inter-Self FAI event; it remains Paper 2's intra-Self territory.

Full-merge as architectural default: The default operation within the shared substrate is full merge of all contributed aspects. Whether full merge is performed for a given event, and which aspects each Self contributes, is human-governed and configurable per event.

Three pattern variants: Full merge of contributed aspects (approximating a union pattern); governance-configured partial merge where human authority selects which contributed content to include (approximating selective merge); full merge with explicit cross-Self provenance carry-over (approximating lineage-preserved union). These three variants are the same three variants Paper 2 establishes for its intra-Self mating primitive, applied at inter-Self scope.

Exchange bounded to substrate content: DNA-layer content (orchestration rules, behavior specifications, schemas) and action-layer content (operational records, lived experience, recorded task instances) exchange through the shared substrate. **LLM weights and instinct-layer content do not exchange.** The boundary is an architectural commitment at the FAI mechanism level, not a governance configuration.

Three persistence loci: Content persists at (a) within the shared substrate during the event, as live coordination medium; (b) within the shared substrate as durable record after dissolution, governance-configured per event; and (c) within each participating Self's home substrate, as evolution outputs ingested through Paper 2's evolution mechanisms.

"Full Aspect Integration" is a working name used in the Paper 3 core theory. The published paper may use a different name for the same architectural commitment. Prior-art notes in this series use this working name with the acknowledgment that it identifies the canonical operation over the shared substrate as Paper 3 specifies it.

3. Aspect as Exchange Unit

The choice of the aspect — rather than the individual cell, or the whole Self — as the unit of exchange is an architectural commitment with specific justification.

Below-aspect granularity (exchanging individual cells without their aspect membership) would surface fragments below their purpose-coherence boundary. Cells in the CKS architecture are the atomic units of operational behavior; they carry meaning in relation to the aspects they compose. Exchanging cells without their aspect context would require the receiving Self to reconstruct purpose-coherence from scratch, defeating the coordination function the exchange is meant to serve.

Above-aspect granularity (exchanging a whole Self) would either require one Self to absorb another's identity — architecturally incoherent given that Selves are governance-distinct entities operating under separate home perimeters — or would amount to merging two Selves into one, a different operation than inter-Self coordination through a shared substrate.

The aspect sits at the right level: it is a purpose-defined arrangement of cells, carrying both the DNA-layer and action-layer content of those cells, with governance structure intact. When a Self contributes an aspect, the shared substrate receives a coherent coordination unit that can be integrated, compared against contributed aspects from other Selves, or carried into evolution outputs, without requiring the receiving context to reconstruct what the contribution means.

This is Paper 2 inheritance. The aspect is defined in Paper 2 as a governed coordination unit grouping cells under a shared purpose. Paper 3 extends the aspect's role from within-Self coordination unit to inter-Self exchange unit without changing its definition. The aspect in FAI is the Paper 2 aspect operating at the inter-Self boundary; it is not a new concept.

The derived sub-commitment is **D1.06**: the aspect as the unit of exchange in FAI, with the claim that below-aspect and above-aspect exchange granularities are distinct architectures that the CKS inter-Self design does not instantiate.

4. N-Ary Cardinality

FAI's cardinality is n-ary as an architectural property: the shared substrate can accommodate contributions from two or more Selves under the same architectural commitments. Two-Self FAI is the simplest instance; three or more Selves participate under the same exchange unit, merge default, pattern variants, and exchange bounding commitments.

The n-ary property matters for prior-art purposes because it forecloses a narrowing interpretation under which FAI applies only to bilateral exchange. A bilateral exchange architecture and an n-ary exchange architecture have different composition properties at population scope — the latter composes directly to Paper 3's Claim 6 (population-scale collective evolution through accumulated FAI events), while the former would require a separate bridging commitment.

The cardinality floor at two reflects Paper 3's scope as addressing inter-Self coordination specifically. One-Self self-derived combination is not an FAI event by definition: it does not cross the inter-Self perimeter, does not involve a shared substrate spanning multiple Selves' home governance perimeters, and produces offspring entities (Paper 2's mating outcome) rather than evolution outputs that each participating Self ingests into its home substrate (FAI's outcome).

The derived sub-commitment is **D1.07**: n-ary cardinality as an architectural property of FAI, with the floor at two marking the inter-Self scope boundary.

5. Full Merge as Architectural Default

The architectural default for FAI is full merge of all contributed aspects within the shared substrate. This default choice has a specific justification: the shared substrate is designed to preserve all contributed content, including conflicts, as first-class substrate state. Full merge as default is consistent with Paper 1's conflict-preservation commitment carrying through the shared substrate Paper 3's Claim 1 establishes. Selective merge as default would require governance to specify what to include before integration can proceed, reversing the preservation-first posture.

The default is configurable. Human authority over the shared substrate includes authority over whether full merge is performed for a given event and which aspects each Self contributes. Governance can select partial merge (Pattern Variant 2) or lineage-preserved union (Pattern Variant 3) for a given event. The architecture does not mandate full merge; it establishes full merge as the posture that applies when governance has not configured otherwise.

This architectural-default commitment is fresh at inter-Self scope. Paper 2's intra-Self mating primitive specifies three pattern variants but does not name a default among them. The naming of full merge as the inter-Self default is a Claim 2 contribution that is not reducible to Paper 2 inheritance, though the pattern variants themselves are inherited.

The derived sub-commitment is **D1.08**: full merge as the architectural default for FAI at inter-Self scope, distinct from the governance-configured variants and not a mandate.

6. Three Pattern Variants

The three pattern variants available under FAI are:

> **Variant 1 — Full merge:** All aspects contributed by all participating Selves are merged within the shared substrate. All contributed content, all conflicts, and all provenance are present in the shared substrate during the event. This is the architectural default.
>
> **Variant 2 — Governance-configured partial merge:** Human authority — operating under joint authority across participating Selves' governance perimeters — configures which contributed content to include in the merge. Content not selected for inclusion may be retained in the shared substrate as record or excluded, per the persistence policy governing the event.

Variant 3 — Full merge with explicit cross-Self provenance carry-over: All contributed aspects are merged (Variant 1), with additional configuration to carry explicit cross-Self provenance references through the shared substrate and into evolution outputs. This variant ensures that evolution outputs ingested by a receiving Self retain traceable lineage back to the contributing Selves' home substrates.

These three variants are the same three variants Paper 2 establishes for its intra-Self combination primitive: union (all parent content combined), selective merge (governance selects which rules to include), and lineage-preserved union (all combined plus explicit cross-lineage references). The correspondence is direct. An adversary cannot claim novelty on any individual FAI variant without bumping into Paper 2's prior art on the corresponding mating variant.

The variants are severable as prior-art sub-claims. Each variant can be defended independently as a named architectural option within the FAI mechanism. Together they constitute the pattern-variant space; individually each is a distinct governance commitment.

The derived sub-commitment is **D1.09**: three FAI pattern variants as severable prior-art sub-claims, with each variant corresponding to its Paper 2 mating analog.

7. Exchange Bounded to DNA-Layer and Action-Layer Content — Instinct-Layer Non-Exchange

The most critical property of FAI for the trilogy's architectural coherence is what does not exchange. **LLM weights and instinct-layer content do not participate in FAI exchange.** This is not a governance configuration that can be overridden; it is an architectural commitment at the FAI mechanism level.

The boundary exists because FAI operates over Paper 2's reasoning layer — the substrate-mediated layer — not over Paper 2's instinct layer, which is implemented in LLM weights and evolves through instinct evolution mechanisms that are architecturally separate from substrate content. The instinct/reasoning separation is Paper 2's foundational claim. FAI inherits that separation and extends it to the inter-Self boundary: the shared substrate is a reasoning-layer artifact, and the content that flows through it is reasoning-layer content.

What this means concretely:

DNA-layer content — orchestration rules, behavior specifications, schemas, governance constraints — can exchange through the shared substrate during an FAI event.

Action-layer content — recorded task instances, operational outputs, decision histories, lived experience accumulated in home substrates — can exchange through the shared substrate during an FAI event.

LLM weights — the parameters of any AI model operating as substrate mediator within any participating Self — do not exchange. They are not substrate content; they are instinct-layer infrastructure.

Instinct-layer content more broadly — any content that belongs to the instinct layer per Paper 2's instinct/reasoning separation — does not exchange. There is no pathway for instinct-layer content to enter the shared substrate through FAI.

This bounding property distinguishes FAI architecturally from federated learning approaches, which operate at parameter or gradient level (instinct-layer content in trilogy terms), and from full-parameter model fusion methods. FAI is not federated learning with aspect wrappers; it is a different kind of operation running at a different layer.

The bounding also closes a specific adversary risk: a party might argue that FAI is a generalization of federated learning by claiming that "aspects" are just named parameter groups. The exchange-bounding commitment forecloses this argument. FAI does not exchange anything at the parameter level; its exchange scope is strictly substrate content.

The derived sub-commitment is **D1.10**: exchange bounded to DNA-layer and action-layer content, with instinct-layer non-exchange as a hard architectural commitment at the FAI mechanism level.

8. Three Persistence Loci

FAI events produce content that persists in three distinct loci, each governance-configurable as a separate object.

Locus 1 — Within the shared substrate during the event. All contributed aspect content, all conflicts surfaced between contributed aspects, all provenance attached to contributed content — this is the live coordination medium for the duration of the FAI event. Its content is the maximal integration state before any persistence policy is applied.

Locus 2 — Within the shared substrate as durable record after dissolution. When the FAI event concludes, the shared substrate dissolves. How much of Locus 1's content is retained as a durable record — ranging from nothing retained, to evolution outputs only, to the full shared substrate retained as auditable record — is governance-configured per event. The persistence policy is itself substrate content, subject to joint human authority across participating Selves' governance perimeters.

Locus 3 — Within each participating Self's home substrate. Evolution outputs — the distillation of the FAI event that each participating Self ingests into its own home substrate through Paper 2's evolution mechanisms — persist within each Self's governance perimeter. The depth of provenance carried with these evolution outputs is governed by the receiving Self's home governance.

The three loci are distinct governance objects. A governance authority that controls Locus 2 configuration does not thereby control Locus 3 configuration; each locus has its own governance perimeter and its own configurability. Collapsing the three loci into a single persistence decision would be an architectural departure from Claim 2.

The derived sub-commitment is **D1.11**: three persistence loci as distinct governance objects, with the configuration of each locus governed separately.

9. FAI as the Inter-Self Analog of Paper 2 Mating

FAI and Paper 2's mating primitive are the same merge primitive at different coordination scopes. Mating is intra-Self: it combines parent entities within one Self's governance perimeter to produce offspring entities. FAI is inter-Self: it combines aspects contributed by distinct Selves through a shared substrate to produce evolution outputs each Self ingests into its home substrate.

The structural correspondence is precise on four dimensions:

> **Exchange unit:** Mating operates on DNA specifications of parent entities; FAI operates on aspects (which carry their constituent cells' DNA-layer and action-layer content). In both cases the unit carries governance structure intact.
>
> **Three pattern variants:** Mating's union, selective merge, and lineage-preserved union correspond directly to FAI's full merge, governance-configured partial merge, and full merge with explicit cross-Self provenance carry-over. The variants are the same architectural options applied at different scopes.
>
> **Human governance:** Both operations are governed by human authority. Mating is governed by the home Self's governance perimeter; FAI is governed by joint authority across participating Selves' governance perimeters.
>
> **Outcome difference:** This is where the analogy breaks. Mating produces offspring entities that persist within the home Self — new cells, new aspects, new governed structures. FAI produces evolution outputs that each participating Self ingests into its home substrate through Paper 2's evolution mechanisms. FAI does not produce inter-Self entities; it produces per-Self evolution material. This outcome difference is what makes FAI a coordination primitive rather than a merger primitive.

The adversary risk this framing closes is that Paper 3's three FAI pattern variants might be argued as novel independent of Paper 2's mating variants. The correspondence is direct: the three FAI variants are not novel at the level of merge posture. What is novel at inter-Self scope is the naming of full merge as the architectural default (Paper 2 names no default among its mating variants) and the application of the conjunction — aspect-level exchange, full-merge default, reasoning-layer bounding — across distinct Selves' home governance perimeters within a temporary shared substrate.

The derived sub-commitment is **D1.12**: FAI as the inter-Self analog of Paper 2 mating, with the outcome difference (evolution outputs vs. offspring entities) as the architectural distinction that places FAI in the inter-Self coordination category rather than the entity-production category.

10. Mapping of Derived Sub-Commitments D1.06–D1.12

Seven sub-commitments derived from Claim 2 will receive independent Phase D1 notes. They are enumerated here so each anchor's scope is recorded at the parent level:

Sub-commitment	Content summary	Phase D1 note
D1.06	Aspect as the unit of exchange in FAI	Forthcoming

D1.07	N-ary cardinality as an architectural property, floor at two	Forthcoming
D1.08	Full merge as architectural default (not mandate) at inter-Self scope	Forthcoming
D1.09	Three FAI pattern variants as severable prior-art sub-claims, corresponding to Paper 2 mating variants	Forthcoming
D1.10	Exchange bounded to DNA-layer and action-layer content; instinct-layer non-exchange as hard architectural commitme	Forthcoming
D1.11	Three persistence loci as distinct governance objects	Forthcoming
D1.12	FAI as inter-Self analog of Paper 2 mating; outcome difference (evolution outputs vs offspring entities)	Forthcoming

Each Phase D1 note will formalize its sub-commitment with full operational detail and boundary-condition analysis. The anchor note (this document) establishes the claim at the paper-claim level; the D1 notes establish each sub-commitment as independent prior art.

11. Operational Test

For an implementation to instantiate FAI as Claim 2 specifies it, an external observer should be able to verify the following:

Test 1 — Exchange unit identification. When a Self contributes content to the shared substrate, can the observer identify the unit of contribution as aspects (not individual cells, not raw content extracted from aspects, not Self-level content)? The shared substrate should contain contributed aspects with their governance structure intact and their constituent cells' DNA-layer and action-layer content surfaced within the aspect boundary.

Test 2 — Layer bounding verification. Does the shared substrate contain any LLM weight content or instinct-layer content from any participating Self? The correct answer is no. If any LLM parameter data, gradient data, or instinct-layer content is present in the shared substrate, the exchange-bounding commitment is violated regardless of how the content arrived there.

Test 3 — Merge default confirmation. In the absence of governance configuration specifying otherwise, does the shared substrate perform full merge of all contributed aspects? The default posture should be full merge with conflict preservation, not selective merge, not refusal to merge pending alignment, not overwrite.

Test 4 — Pattern variant distinguishability. Can the observer determine which of the three pattern variants governs a given FAI event, and can the observer verify that the determination reflects governance configuration under human authority rather than an automatic system choice? If pattern variant selection is opaque or automated without human authority, the governance-configured merge commitment is violated.

Test 5 — Persistence locus separability. Are the three persistence loci — live shared substrate during the event, durable record post-dissolution, and home-substrate evolution outputs per participating Self — governed under separate configuration authorities? The observer should be able to verify that governance configuration for one locus does not automatically determine configuration for another.

An implementation that passes all five tests instantiates FAI at the level of architectural commitment Claim 2 specifies. An implementation that fails any test may still be a valid inter-Self coordination architecture, but it is not the CKS FAI architecture as Paper 3 defines it.

12. Summary

Paper 3's second architectural claim establishes Full Aspect Integration as the canonical operation over the shared substrate its first claim establishes. The claim's content is a conjunction: the aspect (Paper 2 inheritance) as exchange unit; full merge as inter-Self architectural default (fresh at this scope); n-ary cardinality; three pattern variants corresponding to Paper 2's mating variants (inheritance); and exchange bounded to DNA-layer and action-layer content with LLM weights and instinct-layer content excluded by architectural commitment.

The conjunction is the claim. Each component is a severable prior-art sub-commitment (D1.06–D1.12) that this anchor records at the paper-claim level for Phase D1 elaboration.

The exchange-bounding commitment is non-negotiable and non-configurable. It is the mechanism-level expression of the trilogy's foundational instinct/reasoning separation at the inter-Self boundary. An architecture in which instinct-layer content or LLM weights can enter the shared substrate is not a CKS FAI architecture.

FAI's relationship to Paper 2 mating is prior-art framing, not analogy. The three pattern variants are not novel; they are the same three options Paper 2 establishes for its intra-Self primitive, applied at inter-Self scope. What is architecturally fresh is the default commitment and the conjunction at inter-Self scope.

Note #461 in the CKS Derivation Notes series. Parent claim: Paper 3 Claim 2. Derived sub-commitments: D1.06–D1.12 (forthcoming). Prior note: D0.01 (#460). Next note: D0.03 (#462).